

HEMA HASRAN R

AI/ML Engineer | Machine Learning | Deep Learning | Computer Vision

CONTACT

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Location: Coimbatore, Tamil Nadu

PROFESSIONAL SUMMARY

B.Tech student in Artificial Intelligence and Data Science with hands-on experience in Machine Learning, Deep Learning, Computer Vision, and Python-based application development. Skilled in PyTorch, TensorFlow, scikit-learn, OpenCV, and Django, with experience in data preprocessing, feature engineering, model training, evaluation, and predictive analytics.

EDUCATION

B.Tech – Artificial Intelligence and Data Science | 2026

Dr. N.G.P. Institute of Technology, Coimbatore

CGPA: 7.7/10

Diploma in Electrical and Electronics Engineering (EEE) | 2023

Nachimuthu Polytechnic College

Percentage: 85.7%

INTERNSHIP EXPERIENCE

Python Django Developer Intern

Pinesphere Solutions, Coimbatore | July 2024 – August 2024

- Developed backend web applications using Django and PostgreSQL for data-driven solutions.
- Designed and optimized database queries to improve application performance and data management.
- Implemented and tested application features following software development best practices.
- Collaborated with the development team to debug issues and enhance system reliability.

PROJECTS

Human Action Recognition in Videos using Deep Learning

Tech Stack: Python, PyTorch, OpenCV, UCF101

- Developed a deep learning-based video classification system to recognize human activities across 101 UCF101 action categories.
- Implemented video preprocessing, frame extraction, model training, evaluation, and inference workflows using PyTorch and OpenCV.
- Applied computer vision and deep learning techniques to classify human activities from video sequences.

Pneumonia Detection from Chest X-Ray Images

Tech Stack: Python, Scikit-Learn, OpenCV, Random Forest

- Implemented a machine learning-based pneumonia detection system using chest X-ray images.
- Performed image preprocessing and feature extraction to prepare medical image data for classification.
- Trained and evaluated a Random Forest model using accuracy, precision, recall, and F1-score metrics.

Healthcare Disease Prediction System

Tech Stack: Python, Machine Learning, MySQL

- Trained a web-based healthcare disease prediction system using machine learning algorithms.
- Processed healthcare datasets and applied feature engineering techniques for predictive analysis.
- Built an interactive Django interface for disease prediction and visualization of results.

TECHNICAL SKILLS

- Programming:** Python, Java, C, SQL
- AI & Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Predictive Analytics
- Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, NumPy, Pandas
- Development & Databases:** Django, PostgreSQL, MySQL
- Tools:** Git, GitHub, Jupyter Notebook, VS Code, Power BI
- Core Computer Science:** Data Structures & Algorithms, OOP, DBMS

SOFT SKILLS

- Problem Solving, Communication, Teamwork, Adaptability

CERTIFICATIONS & ACHIEVEMENTS

- IBM Data Fundamentals Certification
- IBM Artificial Intelligence Fundamentals Certification
- 1st Place – Programming Quiz & Debugging, Lumino 2K24 Symposium, Bannari Amman Institute of Technology
- Participant – National Level 36-Hour HAC-Kathon 2025, Sri Ramakrishna Engineering College